

ChatSearch Paper Summary

A Dataset and a Generative Retrieval Model for General Conversational Image Retrieval

Multimodal Conversational Image Retrieval | English Summary with Figures



Figure 1: Our generative retrieval model ChatSearcher can accept multimodal inputs and generate textual response or retrieved images. ChatSearcher can comprehend multimodal dialogue context, infer user’s implicit intentions, generate visual or textual responses through multimodal reasoning and world knowledge, and can also support interactive refinement of results.

Figure 1. Examples of ChatSearcher conversational image retrieval interactions.

Executive Insight

ChatSearch shifts image retrieval from single-shot text/image queries into multi-turn multimodal conversations. The model must understand both visual and textual context, infer implicit user intent, and retrieve the desired image from a gallery.

1. Paper Information

Item	Details
Paper Title	ChatSearch: A Dataset and a Generative Retrieval Model for General Conversational Image Retrieval
Main Task	General Conversational Image Retrieval
Main Modalities	Text and Image
Main Output	Textual responses and retrieved image candidates
Main Contribution	A new dataset (ChatSearch) and a generative retrieval model (ChatSearcher)

2. General Idea

The paper introduces a conversational image retrieval setting where a user interacts with a system through multi-turn dialogue. Instead of relying on one direct query, the system uses conversation history, images, and textual instructions to infer the user's hidden retrieval intention.

ChatSearcher can accept interleaved image-text inputs and produce either a textual response or retrieved image candidates. This makes the system more suitable for interactive retrieval scenarios where the user gradually refines the desired image.

3. Problem Statement

Traditional image retrieval systems usually depend on simple interaction forms such as:

- Direct text queries
- Reference images
- Short image descriptions

However, these approaches are limited when the retrieval task involves multi-round multimodal conversations. In such cases, the required information may be implicit and distributed across previous turns, images, and textual feedback.

Core Problem

Existing retrieval models often understand explicit text or image queries, but they struggle to reason over multi-turn multimodal dialogue context and infer implicit user intentions.

4. Proposed Solution

The paper proposes two main components:

1. ChatSearch Dataset: a dataset for general conversational image retrieval based on text-only, image-text, and multi-round multimodal dialogue contexts.
2. ChatSearcher Model: a generative retrieval model trained end-to-end to process interleaved text-image inputs and output textual responses or retrieved images.

5. ChatSearch Dataset Types

The ChatSearch dataset contains three sub-tasks that vary in the type of dialogue context used for retrieval.

Sub-task	Input Context	Goal
tChatSearch	Multi-round textual dialogue	Retrieve image candidates
iChatSearch	Single image + textual instruction	Retrieve image candidates
mChatSearch	Multi-round multimodal dialogue	Retrieve image candidates

	tChatSearch	iChatSearch	mChatSearch
Context modality	text	image-text	image-text
Sample number	5000	10000	10000
Context length	66.7	12.4	56.5
Task input	multi-round textual dialogue	single image and text instruction	multi-round multimodal dialogue
Task output	image candidates	image candidates	image candidates

Table 1: Statistics of ChatSearch test split.

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6. Data Construction Pipeline

The dataset is constructed using foundation models and automatic generation tools. GPT is used to generate dialogue, an image gallery retriever searches candidate images, and an image captioner produces captions. The generated data is then reviewed by human experts for image quality and context relevance.

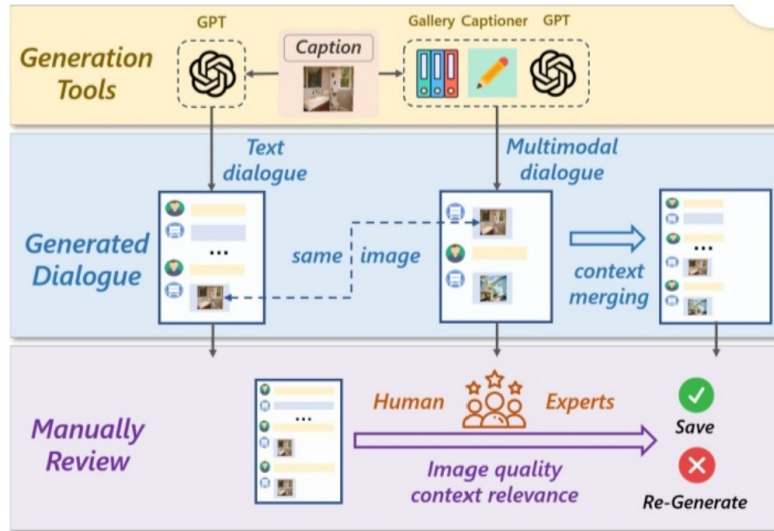
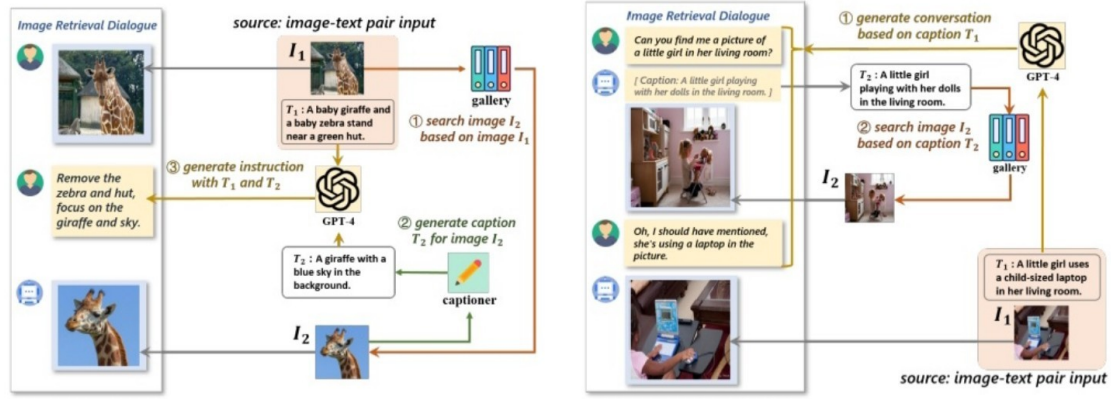


Figure 2: Illustration of automatic data construction pipeline for general conversational image retrieval dataset ChatSearch. We use foundation models (text generator GPT, image gallery retriever and image captioner) as generation tools to generate *text dialogue* and *multimodal dialogue* that aim at searching user desired images, as elaborated in Fig. 3. Then we apply *context merging* method and manually review on those data to construct a high quality evaluation split.

Figure 2. Automatic data construction pipeline for ChatSearch.

The paper also constructs multimodal dialogues using reference images and reference text. This process helps create richer examples where the user's request depends on both visual and textual information.



(a) Dialogue construction with reference image (MDC-I). We use the image I_1 from source image-text pair as the *reference image* to generate a single-round dialogue for image retrieval. (b) Dialogue construction with reference text (MDC-T). We use the text T_1 from source image-text pair as the *reference text* to generate 2-round multimodal dialogue for image retrieval.

Figure 3: Multimodal dialogue construction. The whole pipeline is combined with a text generator, a image search gallery and a pre-trained image captioner. The final output is a user-assistant multimodal conversation designed for image retrieval.

Figure 3. Multimodal dialogue construction using reference image and reference text.

7. ChatSearcher Architecture

ChatSearcher is built as a generative retrieval model. It combines a multimodal large language model with a visual encoder and a retrieval feature queue. The model treats both word prediction and image retrieval as generative processes.

Component	Role
Vicuna-7B LLM	Understands dialogue context and generates textual outputs
CLIP Visual Encoder	Extracts visual features from images
Q-Former	Compresses visual representations before feeding them into the LLM
[IMG] Token	Signals when the model should retrieve an image
Feature Queue	Stores image features as negative/contrastive samples to improve retrieval

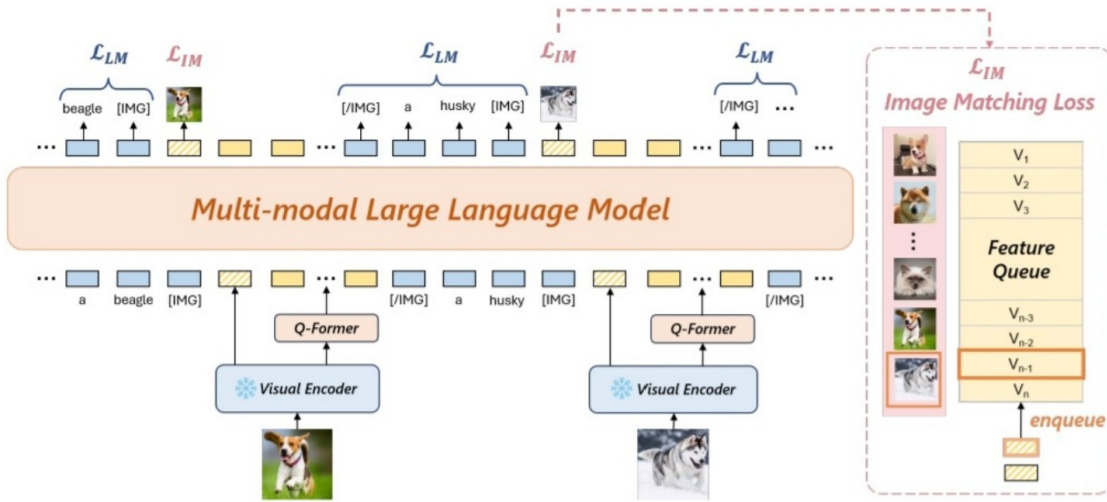


Figure 4: Architecture of our generative retrieval model ChatSearcher. Interleaved documents serve as input, predicting words or retrieving images with generative training objective. Special token `[IMG]` predicts where to retrieve images. We use a dynamically-updated feature queue to save contrastive samples for image retrieval.

Figure 4. ChatSearcher architecture with multimodal LLM, visual encoder, Q-Former, and image matching loss.

8. Training Strategy

Stage	Description
Stage 1: Bidirectional Image-Text Alignment	The model learns image-to-text generation and text-to-image retrieval using interleaved image-text data.
Stage 2: Conversational Instruction Tuning	The model learns to follow multimodal conversational instructions using visual conversation and retrieval-oriented instruction data.

9. Main Results

ChatSearcher outperforms traditional retrieval baselines such as CLIP and FROMAGE on the ChatSearch benchmark. The strongest improvement appears in mChatSearch, which requires understanding multi-turn multimodal dialogue context.

Method	tChatSearch			iChatSearch			mChatSearch			Avg.
	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	
random choice	0.02	0.1	0.2	0.01	0.05	0.1	0.01	0.05	0.1	0.07
CLIP-i [4]	-	-	-	9.65	20.96	28.05	9.65	20.96	28.05	19.55
CLIP-t [4]	15.84	34.46	45.10	14.15	30.60	39.56	12.69	27.19	35.80	28.38
CLIP-ti [4]	-	-	-	12.33	26.81	35.15	13.18	28.56	37.07	25.52
FROMAGE [21]	15.94	36.76	48.60	12.56	29.65	39.65	14.36	32.58	42.80	30.32
ChatSearcher	27.38	52.48	63.50	35.54	61.16	71.57	37.90	64.22	74.06	54.20

Table 2: General conversational image retrieval results on ChatSearch test split.

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Result Interpretation

The results show that conversational history is important. A model that can reason over previous turns and interleaved image-text context retrieves more accurate images than models relying on single-modality or shallow representations.

10. Qualitative Analysis

The qualitative results compare ChatSearcher with CLIP-based retrieval methods. ChatSearcher is better at understanding combined visual and textual clues, while CLIP often fails when the retrieval intention is not explicitly expressed in a single modality.



Figure 7: Qualitative results of ChatSearcher. We show ChatSearcher's conversational image retrieval ability across various dialogue contexts, exhibiting superiority over the vanilla CLIP approaches.

Figure 7. Qualitative comparison between ChatSearcher and CLIP approaches.

11. Research Gap

Although ChatSearch successfully integrates text and images for conversational retrieval, it does not address a broader multimodal setup involving additional signals such as audio, facial expressions, motion, touch, physiological indicators, or environmental sensors.

This creates a research gap for multimodal CIR systems that need to handle richer human-centered signals and real-world conditions. The paper provides a strong foundation for conversational retrieval, but further work is needed to expand beyond text-image interaction.

Extracted Gap

ChatSearch focuses on text-image conversational retrieval. It does not cover audio, facial expressions, motion, touch, physiological signals, or environmental sensors. Therefore, a broader multimodal CIR framework is needed, especially for prototypes that combine audio and facial-expression inputs.

12. Relevance to Our Multimodal CIR Project

The paper is useful for the project because it demonstrates how conversation history and multimodal context can improve retrieval. However, our project can extend this idea toward a more general multimodal framework. While ChatSearch focuses on text and images, our proposed direction considers multiple modalities, and the prototype focuses on audio and facial expressions.

Project Aspect	How ChatSearch Helps
Conversational Retrieval	Shows how multi-turn dialogue can guide retrieval.
Multimodal Reasoning	Demonstrates reasoning over text-image context.
Prototype Direction	Provides a baseline idea that can be extended toward audio-visual emotion-aware retrieval.
Research Gap	Highlights the limitation of text-image-only systems.

13. Key Takeaways

- ChatSearch introduces a new task: general conversational image retrieval.
- ChatSearcher combines multimodal understanding, text generation, and image retrieval.
- Multi-turn multimodal dialogue improves retrieval accuracy compared with single-query approaches.
- The main limitation is that the system focuses only on text and image modalities.
- The paper supports our project by motivating broader multimodal CIR systems that include audio and facial expressions.